

Syn2bANI: chain-restricted maximum-likelihood ANI on fixed restriction-site anchors

a in-silico digestion

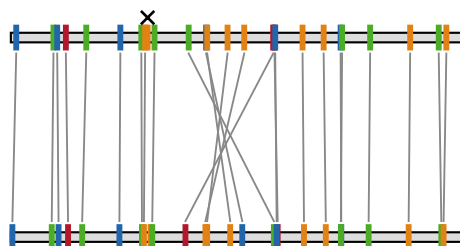
— Bcgl — Alfi — Aloi — Fall

genome A

genome B

positioned Type IIB tags (27–32 bp)

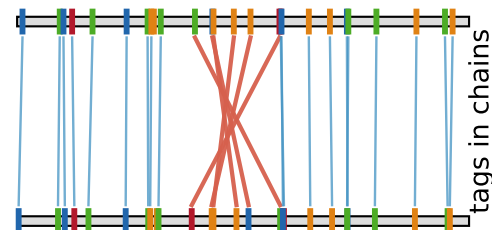
b anchoring (≤ 2 mismatches)



shared tags $\rightarrow$ anchors
(pigeonhole seeding, no scan)

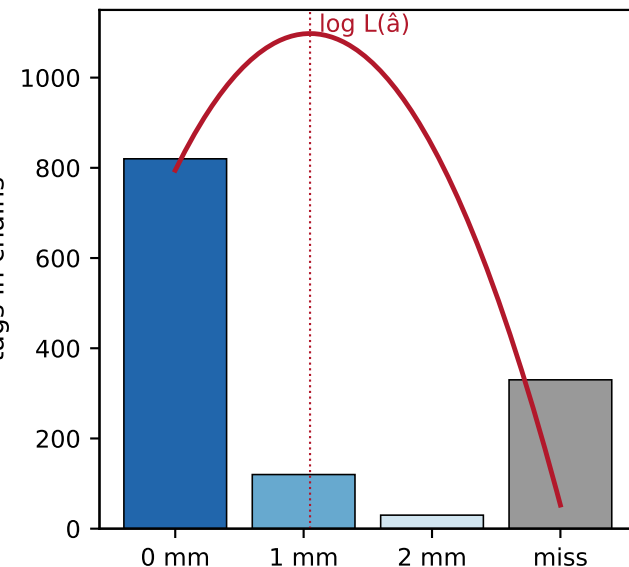
c collinear chaining (2-pass)

— + chains — (inverted) block



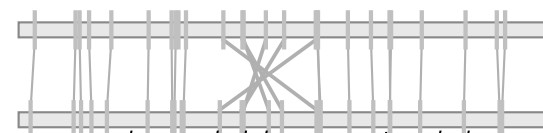
gap-penalized DP;
adaptive chain-break test

d chain-restricted MLE



e k-mer tools (skani / FastANI): same chaining, geometry discarded

k-mer seeds $\rightarrow$ chains (shared step)



k-mer chaining computes chain geometry, then regresses it into a single scalar;
fixed restriction-site anchors retain coordinates, so structure is reported at no extra cost

ANI
97.3

inversions, translocations,
indel positions: not reported

- ANI $\pm$ standard error (gated gamma / uniform MLE)
- aligned fraction af_query / af_reference
- anchor_adjacency, syteny_blocks, breakpoint_count
- SV calls: inversions, indels ≥ 1 kb, translocations
- reliability flag: ok / INCONSISTENT / BELOW_DETECTION
- ani_upper95 when no point estimate is responsible

one sketch, one pass:
~8 ms per pair, 58 MB peak RSS
(vs 8–10 s per pair for dnadiff)